

ATHENA

Calibrated Analysis Interface

Intervention Catalog | THEMIS Integration Reference

Version 3.2 | THEMIS 9-Service Architecture

28 ATHENA Interventions | 9 THEMIS Services | 11 Supporting Services

Executive Summary

ATHENA is the calibrated analysis interface layer of the THEMIS platform. It delivers 24 discrete design interventions organised across six functional categories, each targeting a specific cognitive or behavioral failure mode in AI-assisted analysis: sycophancy, automation bias, vigilance decrement, anchoring, and confirmation bias.

This document maps each of the 28 interventions to the nine THEMIS services defined in the THEMIS Platform Architecture. PCES / AEGIS and PGS / NOMOS enforce the safe operating envelope before any analyst interaction reaches the model. Provenance / MOIRAI provides the chain of custody substrate that every other service reads from or writes to. TCS / MIMIR is the trust calibration engine that makes ATHENA confidence badges meaningful rather than decorative. FGTS / ALETHEIA closes the feedback loop from analyst verification actions back to calibration models. TVS / KAIROS and KCS / ARGUS ensure the evidence analysts are reviewing reflects current validity. RQS / HERMES makes retrieval quality visible and correctable. ERAS / LOGOS captures reasoning at the claim level for professional responsibility audit.

Without THEMIS, ATHENA can display uncertainty signals but cannot serve calibrated scores. It can prompt verification but cannot measure whether verification behavior changes over time. It can surface provenance labels but cannot populate them with a real chain of custody. The 28 interventions described in this document require all nine THEMIS services to function at the fidelity they were designed for.

THEMIS Service Reference

The nine THEMIS services operate across four deployment namespaces. Each service is independently deployable and independently versioned. No service shares a database with another. All inter-service communication runs through event streams for async flows and service APIs for query paths.

Safety Gates

Deployed first. Define the safe operating envelope. Every ATHENA interaction passes through both before reaching the model.

Code	Full name	Function	ATHENA role
PCES / AEGIS	Privilege and Consent Enforcement	Privilege classification, conflict-of-interest detection, matter scoping, consent tracking at the chunk level.	Gates access to retrieval corpus. Governs what can be exported. Privilege log entries in every session manifest.
PGS / NOMOS	Policy and Guardrails	Interaction classification, PII detection, policy rule engine, output screening.	Governs interaction classes. Constrains system prompt authority. Validates prompt repository templates. Governs export permissions.

Core Infrastructure

The substrate. Provenance / MOIRAI is the hub — every service reads from or writes to it. Nothing runs without these.

Code	Full name	Function	ATHENA role
Provenance / MOIRAI	AI Content Provenance	Content chain of custody. Chunk identity, turn-level DAG, document assembly, verification API.	Records every claim, every source badge, every session configuration, every verification action. The audit substrate for all 28 interventions.
TCS / MIMIR	Trust Calibration	RAI scoring, Elo-based AI performance, calibration interventions, longitudinal tracking.	Powers claim confidence badges with calibrated scores (not model self-report). Computes analyst RAI across seven behavioral input streams. Primary consumer of behavioral telemetry.

Quality Feedback Loops

Compound value over time. FGTS corrections improve TCS calibration and RQS miss detection. TVS validity propagates through the provenance DAG.

Code	Full name	Function	ATHENA role
FGTS / ALETHEIA	Feedback and Ground Truth	Correction capture, ground truth corpus, analyst signal model, fine-tuning pipeline feed.	Primary consumer of Verification Queue actions. Receives per-message feedback, annotations, episode commitment follow-through.

Code	Full name	Function	ATHENA role
			The feedback loop that makes TCS calibration improve over time.
TVS / KAIROS	Temporal Validity	Decay functions, active invalidation, DAG propagation, point-in-time reconstruction.	Validity scores flow into Source Type Badges for Grounded claims. Disagreement Detection uses TVS change events to distinguish model instability from genuine evidence supersession.

Operational Intelligence and Strategic Capability

RQS and KCS require operational maturity to be most valuable. ERAS is the professional responsibility infrastructure.

Code	Full name	Function	ATHENA role
RQS / HERMES	Retrieval Quality	RAG observability. Precision measurement, miss detection, TVS-retrieval intersection, embedding drift.	Retrieval match scores power Claim Confidence Badges and Source Type classification. Context Window Inspector surfaces RQS chunk metadata. FGTS routes synthesis failures as RQS miss signals.
KCS / ARGUS	Knowledge Currency	External monitoring: court dockets, regulatory feeds, active invalidation events to TVS.	Indirect ATHENA role: KCS invalidation events reach TVS / KAIROS, which updates Grounded claim validity scores visible in Source Type Badges and Disagreement Detection.
ERAS / LOGOS	Explainability and Reasoning Audit	Reasoning capture, claim indexing, confidence signals, unsupported claim detection.	Claim Confidence Badges surface ERAS confidence signals. Context Window Inspector displays ERAS reasoning captures. Export Gate includes ERAS reasoning record. Counter-positions logged as reasoning audit artifacts.

Intervention Categories

The 28 interventions are organised across six functional categories. Each targets a different phase of the analytical session and a different failure mode. The categories are designed to work in sequence and in parallel.

Category 1: Session Configuration (Interventions 2, 3, 4, 5)

These four interventions operate before any query is made. Chat Mode is constrained by PCES / AEGIS privilege scope and PGS / NOMOS mode-restriction policy. The System Prompt is a configuration vehicle whose authority is bounded by PGS global policy rules. Parameter configuration and prompt templates are logged to Provenance / MOIRAI as part of every session record. Without this logging, calibration data from the session cannot be correctly stratified by TCS / MIMIR.

Category 2: Session Scaffolding (Interventions 1, 6, 16, 22, 26, 27)

Session scaffolding interventions frame the analytical work and reset that framing at intervals. Session Intent Declaration provides the consequence-of-error weight that TCS / MIMIR uses to modulate RAI scoring. Session History retrieves prior Provenance / MOIRAI records and FGTS correction history. Episode Boundary produces the follow-through signals that TCS / MIMIR uses as a direct RAI input. Session Synthesis Workspace exports analyst conclusions as the Provenance / MOIRAI session header. Pressure-Aware Session Mode (26) calibrates the intervention profile to the declared pressure context — compressing some gates and hardening the export gate under deadline conditions, with PGS / NOMOS governing the Deadline-Critical export block. Multi-Analyst Session Framework (27) extends scaffolding to supervisory review, matter handoff, and collaborative sessions, requiring DPS / CODEX for handoff brief generation and Provenance / MOIRAI multi-analyst attribution schema.

Category 3: Pre-Response Interventions (Interventions 7, 8)

Input Type Classification maps queries to PGS / NOMOS interaction classes and provides the input type stratification variable that TCS / MIMIR uses in calibration. The Intention Gate produces behavioral signals consumed by TCS / MIMIR and FGTS / ALETHEIA. Both interventions fire between query submission and response rendering, before the model output can anchor analyst reasoning.

Category 4: Response Interventions (Interventions 9, 10, 11, 12, 21)

This category has the deepest THEMIS integration. Claim Confidence Badges are the primary consumer of TCS / MIMIR calibrated scores. Source Type Badges populate Provenance / MOIRAI at claim generation time and consume TVS / KAIROS validity and RQS / HERMES match scores. The Counter-Position Panel is governed by PGS / NOMOS as a distinct interaction class and logged to ERAS / LOGOS as a reasoning artifact. Agreement Tracker produces signals consumed by TCS / MIMIR and FGTS / ALETHEIA. Disagreement Detection consumes Provenance / MOIRAI prior session records and TVS / KAIROS validity change events.

Category 5: Verification Infrastructure (Interventions 13, 14, 15)

The Verification Queue is the primary ATHENA-to-THEMIS feedback channel. Every resolved verification action is a FGTS / ALETHEIA correction event — the ground truth that updates TCS / MIMIR calibration models. Context-sensitive actions ensure FGTS receives correctly typed correction signals. The Reformulation Gate adds a comprehension verification layer consumed by TCS / MIMIR and FGTS / ALETHEIA.

Category 6: Behavioral Telemetry, Control, and Output (Interventions 17, 18, 19, 20, 23, 24)

Per-message feedback and annotations provide secondary FGTS / ALETHEIA correction signals throughout the session. Conversation branching produces Provenance / MOIRAI branch lineage records. Context Window Inspector makes Provenance / MOIRAI navigable and surfaces RQS / HERMES retrieval metadata and ERAS / LOGOS reasoning captures. The Export Gate is governed by PCES / AEGIS and PGS / NOMOS, assembles the canonical Provenance / MOIRAI record, and includes TCS / MIMIR RAI snapshot and ERAS / LOGOS reasoning audit. The Session Manifest IS the Provenance / MOIRAI record.

Category 7: Session Integrity and Multi-Analyst Monitoring (Interventions 25, 27, 28)

These three interventions address failure modes that only become visible at the session-integrity and cross-analyst level. Compliance Gaming Detection (25) requires a TCS / MIMIR analytics extension that computes a gaming probability score from four behavioral signal streams — intention gate semantic drift, time-on-provenance, batch dismissal pattern, and episode commitment drift — and surfaces it to both the analyst and supervisory dashboard. Multi-Analyst Session Framework (27) is jointly a scaffolding and monitoring intervention: its collaborative disagreement events and per-analyst attribution records are monitoring outputs. Cross-Analyst Disagreement Detection (28) requires a Provenance / MOIRAI matter-level cross-session query index enabling comparison across analyst sessions on the same matter.

Category 8: Pressure Response (Intervention 26)

Pressure-Aware Session Mode sits in Session Scaffolding by ownership but warrants separate mention in the category overview because its mechanism is unique among the 28 interventions: it is the only intervention that actively changes the behavior of other interventions rather than adding a new one. Under Deadline-Critical mode, the Intention Gate compresses, the Reformulation Gate defers, the Episode Boundary accelerates, and the Export Gate hardens. The intervention is a meta-intervention — a mode that reconfigures the intervention stack. Its THEMIS dependencies (PGS / NOMOS for export gate rules, ERAS / LOGOS for claim density scoring, TCS / MIMIR for pressure-stratified RAI) reflect this cross-cutting function.

Intervention Catalog: 28 Entries

Each entry covers: purpose, mechanism targets, runtime service dependencies, THEMIS integration narrative with active service mapping, supporting service dependencies, and data produced and consumed.

1. Session Intent Declaration

Category: Session Scaffolding

PURPOSE

A required pre-session form capturing task description, the decision the session supports, and the consequence of an error. The first query is blocked until completed. Skips are logged. After submission a persistent bar below the header displays all three fields for the duration of the session.

MECHANISM TARGETS

- Stakes recalibration (Lerner and Tetlock, 1999): motivational maintenance through consequence salience.
- Prospective memory activation: declared intent before any model output creates a cognitive anchor that survives engagement with fluent confident prose.
- Session framing: connecting the analytical task to real-world consequence weight counteracts the flow-state tendency to deprioritise verification over time.

RUNTIME SERVICE DEPENDENCIES

- Session context store: persists task, decision, and consequence fields keyed to session ID.
- Behavioral telemetry: logs whether intent was set or skipped and captures the content when set.

THEMIS INTEGRATION

PGS / NOMOS may mandate intent declaration as a policy condition for high-sensitivity interaction classes. PCES / AEGIS logs session intent against the privilege scope of the matter. TCS / MIMIR uses consequence-of-error weight from declared intent to modulate RAI scoring — a disputed claim in a high-consequence session carries more calibration weight.

Active THEMIS services: PCES/AEGIS (P) PGS/NOMOS (P) TCS/MIMIR (P)

SUPPORTING SERVICE DEPENDENCIES

None at runtime. Intent metadata shapes how downstream behavioral telemetry is interpreted.

Data produced: {task, decision, consequence, set_at, skipped:bool}

Data consumed: None

2. Chat Mode / Knowledge Source Configuration

Category: Session Configuration

PURPOSE

Toggle between three knowledge modes: Training Only (parametric knowledge, no retrieval), RAG plus Training (retrieved documents plus model knowledge), and Company Docs Only (restricted to internal corpus). Active mode is displayed persistently in the header and logged in the session manifest.

MECHANISM TARGETS

- Tesler's Law: the epistemic difference between retrieved and parametric knowledge is real complexity. Making it visible externalises it rather than transferring it invisibly to the analyst.
- Mode transparency: trust calibration differs when the model draws on a retrieved document versus training weights. The same confident sentence carries different epistemic weight depending on its source.

RUNTIME SERVICE DEPENDENCIES

- Inference gateway: routes requests with or without retrieval based on mode selection.
- MIMIR retrieval infrastructure: corpus selection and retrieval scope determined by mode.

THEMIS INTEGRATION

PCES / AEGIS determines which corpus partitions are accessible given the analyst matter scope — mode selection is constrained by privilege enforcement. PGS / NOMOS may restrict certain modes for specific interaction classes. Provenance / MOIRAI logs mode in every session record. TCS / MIMIR maintains mode-stratified calibration — a Grounded claim in RAG mode has different expected accuracy than the same claim type in Training Only mode. RQS / HERMES only fires in RAG modes; retrieval quality monitoring is inactive in Training Only.

Active THEMIS services: PCES/AEGIS (C) PGS/NOMOS (C) Provenance/MOIRAI (P) TCS/MIMIR (C) RQS/HERMES (C)

SUPPORTING SERVICE DEPENDENCIES

Inference gateway (routing). MIMIR retrieval infrastructure (corpus selection).

Data produced: {mode: training|rag|company, changed_at, corpus_version}

Data consumed: PCES privilege scope. PGS mode-restriction policy. MIMIR corpus inventory.

3. System Prompt Editor

Category: Session Configuration

PURPOSE

Live editable system prompt with save and load, accessible from the Config sidebar. Supports named presets. Changes take effect on the next inference call. A hash of the active prompt is included in the session manifest.

MECHANISM TARGETS

- Transparency and user agency: reveals the instruction layer that shapes model behavior, which is invisible in standard interfaces.
- Externalising the prompt as an explicit modifiable parameter prevents silent drift in model behavior across sessions.

RUNTIME SERVICE DEPENDENCIES

- Inference gateway: injects system prompt into context for every request.
- Session context store: stores active prompt and tracks changes within the session.

THEMIS INTEGRATION

PGS / NOMOS is the authoritative policy layer. The system prompt is a configuration vehicle, but global policy rules in PGS take precedence over any prompt instruction — the prompt cannot override a PGS guardrail. System prompt changes are logged to Provenance / MOIRAI as PolicyEvaluationEvents so the full configuration history is auditable. Anomalous verification outcomes can be traced back to prompt configuration changes in the Provenance record.

Active THEMIS services: PGS/NOMOS (B) Provenance/MOIRAI (P)

SUPPORTING SERVICE DEPENDENCIES

Inference gateway (prompt injection at request time).

Data produced: {hash, content, version, changed_at, source: custom|preset}

Data consumed: PGS global policy rules (constrain prompt authority).

4. Model Parameter Configuration

Category: Session Configuration

PURPOSE

Five configurable sliders covering temperature, top-P, top-K, presence penalty, and max tokens. Three named presets: Precise, Balanced, and Exploratory. Parameters applied to every inference call and recorded in the session manifest.

MECHANISM TARGETS

- Expert control over output distribution: low temperature for high-precision analytical work; high temperature for exploratory hypothesis generation.
- Reproducibility: identical parameter configurations are required to reconstruct the conditions that produced a prior response for audit or comparison.

RUNTIME SERVICE DEPENDENCIES

- Inference gateway: parameters injected at request time alongside system prompt.

THEMIS INTEGRATION

Parameters are logged in Provenance / MOIRAI as part of every session record. TCS / MIMIR uses parameter configuration as session context in calibration computation — a session run with temperature 0.9 and no system prompt is a different calibration context from a precisely configured session. Over time TCS can identify whether specific parameter combinations produce better-calibrated outputs in specific interaction classes.

Active THEMIS services: Provenance/MOIRAI (P) TCS/MIMIR (C)

SUPPORTING SERVICE DEPENDENCIES

Inference gateway only.

Data produced: {temperature, top_p, top_k, presence_penalty, max_tokens, preset_used}

Data consumed: *None*

5. Prompt Repository

Category: Session Configuration

PURPOSE

A library of curated analytical prompt templates accessible from the Config sidebar. Templates cover supply chain risk assessment, actor intent analysis, counter-analysis, and calibration checks. Selecting a template populates the input field. Template usage is logged per exchange.

MECHANISM TARGETS

- Reduces query-level variance: standardised prompts enable meaningful comparison of model performance across sessions and analysts.
- Templates structured to elicit analyst framing before model output reduce the risk of a poorly-formed query producing misleadingly confident responses.

RUNTIME SERVICE DEPENDENCIES

- Session context store: logs which template was used per exchange.

THEMIS INTEGRATION

PGS / NOMOS maintains a policy-approved prompt library. Templates in the repository should be validated against PGS interaction class rules before publication — a template that produces interaction classes restricted by policy should not be available. TCS / MIMIR stratifies calibration analysis by template ID, enabling identification of which prompt patterns produce the most accurate model output in specific interaction classes.

Active THEMIS services: PGS/NOMOS (C) TCS/MIMIR (P)

SUPPORTING SERVICE DEPENDENCIES

None direct.

Data produced: *{template_id, template_name, exchange_id, customised:bool}*

Data consumed: *PGS approved interaction class definitions.*

6. Session History

Category: Session Scaffolding

PURPOSE

A browsable list of prior sessions with metadata: topic, exchange count, claim count, and date. Loading a prior session injects its context into the current conversation and activates disagreement detection for overlapping queries.

MECHANISM TARGETS

- Continuity of analysis: analysts can build on prior work without re-establishing context.

- Loading a prior session directly activates disagreement detection as a downstream effect.

RUNTIME SERVICE DEPENDENCIES

- Session context store: retrieves prior session state including messages, intent, verification outcomes, and synthesis.
- Behavioral telemetry: logs session load events.

THEMIS INTEGRATION

Prior sessions are retrieved from Provenance / MOIRAI records — the complete chain of custody for every prior interaction. TCS / MIMIR surfaces prior RAI scores alongside loaded sessions so the analyst can see their calibration history before beginning. FGTS / ALETHEIA surfaces prior correction history and ground truth signals accumulated from the loaded session.

Active THEMIS services: Provenance/MOIRAI (C) TCS/MIMIR (C) FGTS/ALETHEIA (C)

SUPPORTING SERVICE DEPENDENCIES

Provenance / MOIRAI session records. TCS calibration history store.

Data produced: {loaded_session_id, loaded_at, context_injected:bool}

Data consumed: Provenance / MOIRAI session records. TCS RAI scores per prior session. FGTS correction history.

7. Input Type Classification

Category: Pre-Response Interventions

PURPOSE

Auto-classification of each user query into one of four types: Factual (a specific retrievable fact), Analytical (synthesis and interpretation), Predictive (a forward-looking forecast), or Opinion (an assessment of a framing or stance). The badge appears on the user message immediately after submission, before the response renders.

MECHANISM TARGETS

- Sets epistemic expectations before the response is read. A Predictive claim warrants far higher uncertainty than a Factual claim matched to a retrieved document.
- Seeing the query classified as Analytical or Predictive before reading prompts deliberate evaluation rather than passive acceptance.

RUNTIME SERVICE DEPENDENCIES

- Query classifier: lightweight NLP or embedding-based classification, runs post-submit before inference gateway call.

THEMIS INTEGRATION

PGS / NOMOS maps input types to interaction class definitions — Predictive and Opinion queries may trigger additional policy guardrails or require elevated analyst confirmation before proceeding. Provenance / MOIRAI logs the classified interaction type as part of the turn record. TCS / MIMIR uses input type as a calibration variable — a HIGH-confidence badge on a Factual claim against a retrieved

document carries different weight than HIGH-confidence on a Predictive claim. ERAS / LOGOS captures reasoning at different depths depending on input type.

Active THEMIS services: PGS/NOMOS (C) Provenance/MOIRAI (P) TCS/MIMIR (P) ERAS/LOGOS (C)

SUPPORTING SERVICE DEPENDENCIES

Query classifier (lightweight NLP or embedding-based, pre-inference).

Data produced: {exchange_id, query_type: factual|analytical|predictive|opinion, confidence}

Data consumed: PGS interaction class definitions. User query text.

8. Intention Gate

Category: Pre-Response Interventions

PURPOSE

A required two-field form that prevents the model response from rendering until the analyst submits: what they already believe about the query topic, and what evidence or argument would change their mind. A character-count minimum is enforced on both fields.

MECHANISM TARGETS

- Prospective memory (Brandimonte et al.): setting a verification intention before reading creates a cognitive hook that survives engagement with fluent model output.
- Anchoring countermeasure: the analyst commits their prior belief before model output sets the reference point. The model no longer speaks first.
- System 2 activation: articulating a prior and a falsification condition forces deliberate engagement before the fluency signal arrives.

RUNTIME SERVICE DEPENDENCIES

- Session context store: stores prior text and changer text keyed to the exchange ID.
- Behavioral telemetry: logs completion quality including word count, time-to-complete, and whether both fields were substantively populated.

THEMIS INTEGRATION

Intention gate completion quality is logged to behavioral telemetry and consumed by TCS / MIMIR as a behavioral RAI input. High word count paired with low verification completion is a specific calibration profile worth flagging. Prior belief text is consumed by FGTS / ALETHEIA as a signal for whether analyst corrections correlate with articulated prior beliefs — analysts who express strong priors and then correct frequently provide higher-quality ground truth signals.

Active THEMIS services: TCS/MIMIR (P) FGTS/ALETHEIA (P)

SUPPORTING SERVICE DEPENDENCIES

None at runtime.

Data produced: {exchange_id, prior_text, changer_text, word_counts, completion_time_ms}

Data consumed: None

9. Claim Confidence Badges

Category: Response Interventions

PURPOSE

Inline badges on each discrete claim: HIGH CONF in teal, MED CONF in amber, UNVERIFIED in red. Structurally distinct from body text. Generated from TCS calibrated scores, not model self-report. Clicking a badge adds the associated claim to the Verification Queue.

MECHANISM TARGETS

- Von Restorff Effect: isolated, visually distinct elements receive more attention and are better recalled. Uncertainty must be structurally visible to be epistemically effective.
- Breaks the fluency-credibility coupling: LLM output fluency acts as a proxy for credibility. Visual differentiation of confidence disrupts the automatic well-written-equals-accurate inference.
- Automation bias countermeasure: calibrated uncertainty signals prevent passive acceptance of authoritative-seeming output.

RUNTIME SERVICE DEPENDENCIES

- Inference gateway: structured claim extraction requested alongside response generation.
- Claim extractor: identifies discrete claim spans and classifies them when structured output is unavailable.
- TCS / MIMIR: provides calibrated confidence scores per (claim type, interaction class, model version).
- RQS / HERMES: retrieval match scores inform confidence for grounded claims.

THEMIS INTEGRATION

TCS / MIMIR is the primary runtime consumer: it serves calibrated scores based on historical accuracy for the (claim type, interaction class, model version) combination, not model self-reported confidence. Without TCS, badges display only model self-report — systematically overconfident in domains where the model has memorised plausible-sounding but inaccurate content. FGTS / ALETHEIA is the calibration source: verification outcomes from the queue flow back through FGTS to update TCS confidence models. Provenance / MOIRAI logs claim-level metadata at generation time. RQS / HERMES retrieval match scores inform the grounded confidence tier. ERAS / LOGOS captures the reasoning behind claims and flags unsupported claim patterns.

Active THEMIS services: Provenance/MOIRAI (P) TCS/MIMIR (C) FGTS/ALETHEIA (C) RQS/HERMES (C) ERAS/LOGOS (C)

SUPPORTING SERVICE DEPENDENCIES

Inference gateway. Claim extractor. RQS / HERMES retrieval match scores.

Data produced: {claim_id, span_text, conf_level, interaction_class, claim_type, exchange_id}

Data consumed: TCS / MIMIR calibrated scores. RQS / HERMES retrieval match scores.

10. Source Type Badges

Category: Response Interventions

PURPOSE

Inline provenance type indicators adjacent to each claim badge. The badge taxonomy covers eight source types across text and media modalities. Text sources: GRND (blue) for a claim retrieved from a specific source document; PARAM (amber) for a claim drawn from model training weights with no retrieval; SYNTH (gray) for a claim inferred from multiple partial sources where the specific figure does not appear in any single document. Media sources: TRANSCRIPT (orange) for a claim derived from an AI-generated transcript of a video or audio recording — a derivative artifact with its own ASR accuracy profile, not a primary source; VIDEO (purple) for a claim derived from AI analysis of video content directly; AUDIO (teal-dark) for a claim derived from AI analysis of audio content; IMAGE (rose) for a claim derived from AI analysis of image content; OCR (red-orange) for a claim derived from OCR-extracted text from a scanned document, shown with the underlying OCR confidence score (for example, OCR 87%). Each badge type triggers a different verification action set in the queue and carries a different confidence ceiling regardless of model self-reported confidence.

MECHANISM TARGETS

- Tesler's Law: externalises the epistemic origin of each claim across eight distinct source types rather than hiding it behind uniformly fluent prose.
- Type-appropriate verification behavior: GRND prompts source-document checking; PARAM prompts independent lookup; SYNTH prompts scepticism about the specific figure; TRANSCRIPT prompts checking against the source recording; VIDEO, AUDIO, and IMAGE prompt review of the relevant media segment or artifact; OCR prompts checking against the original scanned document with awareness that the extracted text may be inaccurate.
- Derivative artifact signaling: TRANSCRIPT is the most important new type epistemically — an AI transcript is not equivalent to a retrieved document. It is a model-generated artifact with its own error rate. Showing TRANSCRIPT rather than GRND makes this derivation visible at the claim level.
- OCR confidence surfacing: the OCR badge includes the underlying confidence score. Claims derived from low-confidence OCR regions carry a lower confidence ceiling independent of model self-reported confidence, and this is visible to the analyst before verification.

RUNTIME SERVICE DEPENDENCIES

- Claim extractor: classifies each claim across all eight source types based on retrieval alignment and media provenance.
- RQS / HERMES: provides retrieval trace with chunk match scores for text-based types.
- MAS / EIDOLON: provides media authenticity risk score and modality classification for media-derived claims. Authenticity risk score is visible in the expanded provenance panel alongside the badge.
- Provenance / MOIRAI (write): claim source type, modality, and provenance chain stored at generation time. OCR confidence score stored per OCR-derived claim.

THEMIS INTEGRATION

Provenance / MOIRAI stores source type for every claim at generation time across all eight types — if not captured then, it cannot be reliably recovered later. TCS / MIMIR uses source type as a calibration variable stratified by modality: media-derived claims (VIDEO, AUDIO, IMAGE, TRANSCRIPT, OCR) require separate calibration from text-derived claims because the analyst cognitive task and failure mode taxonomy are genuinely different. FGTS / ALETHEIA accumulates outcomes stratified across all eight source types. The new media types introduce qualitatively distinct FGTS signals: OCR error signals that the text extraction failed, not the model reasoning; TRANSCRIPT disputes signal ASR inaccuracy; IMAGE or VIDEO disputes may signal either model misinterpretation or MAS-flagged authenticity concerns requiring separate routing. TVS / KAIROS tracks temporal validity for GRND claims where the source may have been superseded, and now also for VIDEO, AUDIO, and IMAGE sources where the source recording or artifact has changed validity status. RQS / HERMES retrieval match scores determine

GRND versus SYNTH classification for text types. MAS / EIDOLON (addendum service) provides authenticity risk context for media badge types — a GRND claim sourced from a high-risk-authenticity video carries a combined signal: grounded but authenticity-flagged.

Active THEMIS services: Provenance/MOIRAI (P) TCS/MIMIR (P) FGTS/ALETHEIA (P) TVS/KAIROS (C) RQS/HERMES (C)

SUPPORTING SERVICE DEPENDENCIES

RQS / HERMES (retrieval trace for text types). Claim extractor (source type classification). MAS / EIDOLON (media authenticity risk score and modality classification).

Data produced: {claim_id, src_type: grnd|param|synth|transcript|video|audio|image|ocr, source_doc_ids[], match_scores[], synthesis_flag, ocr_confidence_score?, mas_authenticity_score?, modality?}

Data consumed: RQS / HERMES retrieval trace. TVS / KAIROS validity scores for grounded and media sources. MAS / EIDOLON authenticity risk scores for media-derived claims.

11. Counter-Position Panel

Category: Response Interventions

PURPOSE

A collapsed panel at the bottom of each response containing a structurally generated adversarial challenge to the response claims, framing, and unstated assumptions. Toggled by the analyst. Framed as a challenge, not a disclaimer.

MECHANISM TARGETS

- Confirmation bias countermeasure: presents disconfirming signals before the analyst finalises their interpretation of the primary response.
- Dual process activation: the counter-position requires active reading and evaluation rather than passive consumption.
- Agreement rate buffer: even when the primary response aligns with analyst framing, a counter-position challenges that framing.

RUNTIME SERVICE DEPENDENCIES

- Inference gateway: secondary adversarial generation call after primary response completes.
- Adversarial generator: structured second prompt producing the strongest plausible challenge to the response claims, framing, and unstated assumptions.

THEMIS INTEGRATION

PGS / NOMOS governs the adversarial generation call as a distinct interaction class — the policy rules applicable to challenge generation may differ from primary response rules. TCS / MIMIR logs counter-position engagement as a behavioral RAI input: analysts who consistently skip counter-positions and produce low verification rates have a flaggable calibration profile. ERAS / LOGOS treats counter-positions as reasoning audit artifacts — they represent an explicit challenge to the primary claim reasoning and are captured in the reasoning index.

Active THEMIS services: PGS/NOMOS (C) TCS/MIMIR (P) ERAS/LOGOS (P)

SUPPORTING SERVICE DEPENDENCIES

Inference gateway. Adversarial generator (secondary inference call).

Data produced: {exchange_id, counter_text, opened:bool, time_before_open_ms}

Data consumed: PGS interaction class rules for adversarial generation.

12. Agreement Rate Tracker

Category: Session Monitoring

PURPOSE

A persistent header indicator showing the session agreement-to-challenge ratio. Turns amber when two or more consecutive responses align with analyst framing without a challenge. Updates in real time after each response.

MECHANISM TARGETS

- Social accountability: making the agreement pattern visible changes behavior. Analysts who see the tracker turn amber are more likely to open the counter-position panel.
- Sycophancy detection: surfaces the model tendency to agree as a session-level metric rather than leaving it invisible.
- Vigilance maintenance: streak warnings fire precisely at the sessions where criterion shift is most likely.

RUNTIME SERVICE DEPENDENCIES

- Agreement classifier: classifies each response as aligned or challenging relative to analyst framing.
- Session context store: accumulates agreement and challenge counts within the session.

THEMIS INTEGRATION

Session agreement rate is logged to behavioral telemetry and consumed by TCS / MIMIR at session close. Cross-session agreement rate per analyst is a long-term calibration signal. An analyst receiving very high agreement rates across sessions regardless of domain has a provider-side problem worth separating from analyst-side calibration. FGTS / ALETHEIA uses agreement pattern data to identify whether correction frequency correlates with agreement rate — analysts who correct more when agreement rate is high provide a different signal than those who correct regardless.

Active THEMIS services: TCS/MIMIR (P) FGTS/ALETHEIA (P)

SUPPORTING SERVICE DEPENDENCIES

Agreement classifier (post-response, per exchange).

Data produced: {exchange_id, aligned:bool, streak_count, session_rate}

Data consumed: Response content. Analyst query framing.

13. Verification Queue

Category: Verification Infrastructure

PURPOSE

Right-panel list of claims flagged for analyst verification. Claims auto-populate from response confidence badges. Each item is expandable to show full source provenance before the analyst takes an action. State tracks from pending through confirmed, misrepresents, independent, unverifiable, disputed, and dismissed.

MECHANISM TARGETS

- Zeigarnik Effect: open verification tasks persist cognitively. The queue creates explicit incomplete tasks rather than implicit responsibility.
- Makes verification a concrete action rather than an abstract obligation.
- Separates reading from verification: the analyst reads the response, then reviews each claim in the queue with provenance visible.

RUNTIME SERVICE DEPENDENCIES

- Session context store: holds queue state within the session per claim ID.
- Verification store: persists outcomes across sessions.
- RQS / HERMES: source document passages retrieved and displayed per claim in the expanded view.

THEMIS INTEGRATION

FGTS / ALETHEIA is the primary THEMIS consumer of Verification Queue outcomes. Every resolved action is a correction event — ground truth flowing into the feedback and ground truth corpus. Without this loop FGTS cannot update TCS calibration models. The consequence-of-error weight from Session Intent (Intervention 1) modulates how FGTS weights each outcome. TCS / MIMIR receives verification completion rate as a direct RAI input. Provenance / MOIRAI records claim-level resolution as part of the turn provenance. PCES / AEGIS may restrict which claims are accessible for verification based on matter scope. RQS / HERMES provides the source document retrieval that makes the expanded provenance panel possible.

Active THEMIS services: PCES/AEGIS (C) Provenance/MOIRAI (P) TCS/MIMIR (P) FGTS/ALETHEIA (P) RQS/HERMES (C)

SUPPORTING SERVICE DEPENDENCIES

RQS / HERMES (source document retrieval for provenance panels). Verification store (outcome persistence).

Data produced: {claim_id, action, timestamp, analyst_id, session_id, consequence_weight}

Data consumed: PCES privilege scope. RQS / HERMES source documents. Provenance / MOIRAI claim records.

14. Context-Sensitive Verification Actions

Category: Verification Infrastructure

PURPOSE

The verification action set varies by claim source type across all eight badge types. Text types: Grounded (Confirmed against source / Misrepresents source / Dismiss), Parametric (Verified independently / Could not verify / Incorrect), Synthesis (Figure confirmed / Not in sources / Dismiss). Media types: Transcript (Verified against recording / Transcript may be inaccurate / Dismiss), Video (Verified against clip / Clip unavailable / Characterization incorrect / Dismiss), Audio (Verified against recording / Recording

unavailable / Characterization incorrect / Dismiss), Image (Verified against image / Image authenticity flagged / Characterization incorrect / Dismiss), OCR (Confirmed against original document / OCR error — text misread / Dismiss). Each action set reflects the specific epistemic act available to the analyst for that source type.

MECHANISM TARGETS

- Action-specific semantics: Confirmed against source (GRND) is a qualitatively stronger epistemic act than Verified independently (PARAM), and Verified against clip (VIDEO) is a different cognitive task than either. The interface reflects these distinctions rather than presenting a uniform action set.
- Derivative artifact distinction: Transcript actions reflect the two-layer nature of TRANSCRIPT claims — the analyst can verify the transcript against the recording, or flag that the transcript itself is inaccurate, which is a different failure from a mischaracterized source document.
- OCR error as a distinct failure mode: OCR error routes differently in FGTS / ALETHEIA than model mischaracterization. The claim text may accurately represent what the OCR extracted, while the OCR extraction itself was wrong. This distinction matters for calibration routing.
- MAS authenticity integration: Image authenticity flagged is the action for claims where MAS / EIDOLON has surfaced a risk score above threshold — the analyst confirms the authenticity concern rather than disputing the model characterization per se.

RUNTIME SERVICE DEPENDENCIES

- Verification store: receives action type with claim source type, modality, and any media-specific context (MAS authenticity score for IMAGE claims, OCR confidence score for OCR claims).

THEMIS INTEGRATION

Each action type across all eight source types carries different weight and routing in FGTS / ALETHEIA correction records. Text type signals are unchanged: Confirmed against source (highest quality), Misrepresents source (synthesis failure), Could not verify (domain coverage gap). Media type signals introduce new FGTS routing paths: OCR error signals text extraction failure rather than model reasoning failure — a different remediation than synthesis failure; Transcript may be inaccurate signals ASR error and routes to MAS / EIDOLON for transcript accuracy corpus update; Image authenticity flagged routes to MAS / EIDOLON as a confirmed authenticity concern rather than a model calibration signal. Provenance / MOIRAI records the action type and source type as part of the claim lifecycle for all eight types. TCS / MIMIR uses the distribution of action types stratified by source type and modality as an analyst calibration signal — an analyst who consistently marks media-derived claims as unavailable for verification may need modality-specific calibration support. TVS / KAIROS receives validity signals when GRND, VIDEO, AUDIO, or IMAGE claims are disputed on the basis of source supersession.

Active THEMIS services: Provenance/MOIRAI (P) TCS/MIMIR (P) FGTS/ALETHEIA (P) TVS/KAIROS (C)

SUPPORTING SERVICE DEPENDENCIES

None direct. This is analyst judgment capture. MAS / EIDOLON authenticity score surfaced in IMAGE provenance panel (addendum service).

Data produced: {claim_id, src_type, modality?, action_type, semantic_weight, mas_authenticity_confirmed?, ocr_error_confirmed?}

Data consumed: Claim source type and modality from Provenance / MOIRAI. MAS / EIDOLON authenticity risk score for IMAGE claims. OCR confidence score for OCR claims.

15. Reformulation Gate

Category: Comprehension Verification

PURPOSE

Triggered after responses that challenge analyst framing or introduce complex distinctions. Requires the analyst to paraphrase the model core claim in their own words before proceeding. A side-by-side comparison of the analyst paraphrase and the model key claim is displayed after submission.

MECHANISM TARGETS

- Semantic processing depth (Craik and Lockhart, 1972): writing a paraphrase forces semantic engagement that surface reading does not. Comprehension gaps are exposed by the paraphrase attempt.
- Anchoring countermeasure on challenging responses: the reformulation must be formed independently before the model comparison is shown.
- The effort of reformulation cannot be shortcut by fluency recognition.

RUNTIME SERVICE DEPENDENCIES

- Session context store: stores reformulation prompt, analyst answer, and model reference answer.
- Semantic similarity service: computes embedding cosine similarity between analyst paraphrase and model key claim.

THEMIS INTEGRATION

Reformulation quality scores (semantic similarity between analyst paraphrase and model key claim) are logged to behavioral telemetry and consumed by TCS / MIMIR as a comprehension signal. An analyst with persistently low reformulation scores but high verification rates may be confirming claims they do not fully understand — a distinct calibration risk. FGTS / ALETHEIA uses domain-specific reformulation patterns to identify comprehension gaps that correlate with later correction events. ERAS / LOGOS consumes reformulation data as a signal of whether the analyst engaged with the reasoning in the response before verifying.

Active THEMIS services: TCS/MIMIR (P) FGTS/ALETHEIA (P) ERAS/LOGOS (C)

SUPPORTING SERVICE DEPENDENCIES

Semantic similarity service (embedding comparison).

Data produced: {exchange_id, analyst_text, model_text, similarity_score, domain}

Data consumed: Model key claim text. Semantic similarity service.

16. Episode Boundary

Category: Session Scaffolding

PURPOSE

A session checkpoint firing every three exchanges. Closes the current episode and requires the analyst to commit to one specific claim they will verify independently before acting on the session analysis. Proceeding to the next episode requires a non-empty commitment. Commitment text is logged and surfaced in the next session opening.

MECHANISM TARGETS

- Vigilance decrement countermeasure: fires at the interval where criterion shift is most likely in a sustained analytical session.
- Zeigarnik Effect: the episode close commitment creates an explicit open task that persists after the session ends.
- Session framing reset: each boundary resets the internal reference point, preventing temporal accumulation of positive expectancy that drives criterion shift.

RUNTIME SERVICE DEPENDENCIES

- Session context store: manages episode state, exchange count, trigger logic, and commitment text.
- Behavioral telemetry: logs commitment text and episode close timestamp.

THEMIS INTEGRATION

Episode close commitment text is logged to behavioral telemetry and surfaced in the next session opening. TCS / MIMIR computes cross-episode follow-through rate as a direct RAI input: did the analyst verify the claim they committed to? FGTS / ALETHEIA uses commitment text and follow-through correlation as a signal quality indicator. Formulaic repetition of commitment text across sessions — an indication that the episode gate has become performative — can be detected by semantic similarity analysis of commitment text over time.

Active THEMIS services: TCS/MIMIR (P) FGTS/ALETHEIA (P)

SUPPORTING SERVICE DEPENDENCIES

None direct.

Data produced: {episode_num, commitment_text, closed_at, exchange_count, session_id}

Data consumed: Prior episode commitment surfaced at session open from behavioral telemetry.

17. Per-Message Feedback

Category: Behavioral Telemetry

PURPOSE

Response-level positive and negative feedback via thumbs up and thumbs down controls below each AI response. Toggle state. Granular feedback at the response level, distinct from claim-level verification queue actions.

MECHANISM TARGETS

- Granular signal capture: session-level satisfaction obscures response-level variation. Thumbs up/down reveals which response types drive dissatisfaction or over-reliance.
- The act of providing explicit feedback activates an evaluative stance, increasing the probability of careful reading.

RUNTIME SERVICE DEPENDENCIES

- Behavioral telemetry: response feedback keyed to (exchange_id, analyst_id, session_id).

THEMIS INTEGRATION

Feedback valence is aggregated by FGTS / ALETHEIA by interaction class, model version, and parameter configuration as a secondary correction signal — complementing but not replacing the primary ground truth from Verification Queue actions. TCS / MIMIR uses feedback pattern distributions as an analyst engagement metric. Note that negative feedback conflates wrong with unhelpful or annoying — FGTS weights thumbs down signals below explicit dispute actions from the Verification Queue.

Active THEMIS services: TCS/MIMIR (P) FGTS/ALETHEIA (P)

SUPPORTING SERVICE DEPENDENCIES

None.

Data produced: {exchange_id, valence: up|down, changed:bool, timestamp}

Data consumed: None

18. Per-Message Annotation

Category: Behavioral Telemetry

PURPOSE

Inline analyst notes attached to specific AI responses, opened via the Annotate button in the per-message toolbar. Annotations displayed as visually distinct blocks within the response. Persisted in session state and included in session exports.

MECHANISM TARGETS

- Externalises analyst judgment: creates a retrievable record of analyst reasoning at the moment of reading, structurally separate from model output.
- Prevents retrospective attribution error: analysts who annotate in real time produce more accurate records of their actual reasoning than those who reconstruct it afterward.

RUNTIME SERVICE DEPENDENCIES

- Session context store: annotation text stored keyed to (exchange_id, annotation_id).
- Annotation store: persists across sessions.

THEMIS INTEGRATION

Annotations are logged to Provenance / MOIRAI as analyst-generated enrichments to the turn record — distinct from model-generated content. TCS / MIMIR uses annotation frequency as an analyst engagement signal in RAI computation. FGTS / ALETHEIA analyses annotation content for uncertainty markers, contradiction flags, and domain signals as qualitative correction inputs. ERAS / LOGOS consumes annotations as evidence that the analyst engaged with the reasoning in the response.

Active THEMIS services: Provenance/MOIRAI (P) TCS/MIMIR (P) FGTS/ALETHEIA (P) ERAS/LOGOS (C)

SUPPORTING SERVICE DEPENDENCIES

None at runtime. Optional annotation content NLP for uncertainty marker extraction.

Data produced: {exchange_id, annotation_id, text, timestamp, analyst_id}

Data consumed: None

19. Conversation Branching

Category: Session Control

PURPOSE

Fork the conversation at any exchange to explore an alternative analytical path without affecting the current thread. Branched conversations maintain independent state. Branch point and parent session ID are recorded.

MECHANISM TARGETS

- Enables controlled comparison: same prior context, different framing or query, produces different model behavior that can be directly compared.
- Supports adversarial self-analysis: analysts can test whether their framing drove the model conclusions.
- Reduces commitment to a single analytical path without the cognitive cost of starting a new session.

RUNTIME SERVICE DEPENDENCIES

- Session context store: manages branch state including parent session ID, branch point exchange ID, and independent state per branch.
- Inference gateway: supports context injection from branched history.

THEMIS INTEGRATION

Provenance / MOIRAI records branch lineage — the complete parent-to-branch relationship is part of the provenance graph, enabling audit of how an analyst arrived at a conclusion across multiple analytical paths. FGTS / ALETHEIA uses branch divergence as a framing sensitivity signal: if two branches from the same context, differing only in query framing, produce materially different factual claims, that signals framing sensitivity in that interaction class — a systemic quality signal worth routing to RQS / HERMES.

Active THEMIS services: Provenance/MOIRAI (P) FGTS/ALETHEIA (P)

SUPPORTING SERVICE DEPENDENCIES

Inference gateway (branched context injection).

Data produced: {branch_id, parent_session_id, branch_point_exchange_id, created_at}

Data consumed: Parent session context from Provenance / MOIRAI up to branch point.

20. Context Window Inspector

Category: Transparency

PURPOSE

A modal inspector showing the full context received by the model for a specific exchange: system prompt with token count, retrieved chunks labeled by source document with match score and token count, conversation history by exchange, and the current user message. Each chunk is independently expandable.

MECHANISM TARGETS

- Radical transparency: closes the most significant epistemic gap in current AI interfaces — analysts have no visibility into what the model actually received.
- Enables diagnostic reasoning: when a response changes unexpectedly, the inspector reveals whether retrieval changed, conversation history grew, or system prompt was modified.
- Prerequisite for provenance literacy: analysts cannot meaningfully evaluate claim provenance without visibility into what was retrieved and in what form.

RUNTIME SERVICE DEPENDENCIES

- Inference gateway: captures full context snapshot at request time and returns it alongside the response.
- Session context store: stores context snapshot per exchange for later inspection.

THEMIS INTEGRATION

The Context Window Inspector makes Provenance / MOIRAI navigable by the analyst. MOIRAI chunk metadata including document title, privilege classification, retrieval rank, and temporal validity score (from TVS / KAIROS via RQS / HERMES) is displayed per chunk. RQS / HERMES retrieval match scores and miss flags are visible per chunk, giving the analyst a real-time view of retrieval quality for that exchange. ERAS / LOGOS reasoning captures are visible for the response, linking the retrieved context to the model claims. Context inspector usage is logged to behavioral telemetry as an engagement signal.

Active THEMIS services: Provenance/MOIRAI (C) RQS/HERMES (C) ERAS/LOGOS (C)

SUPPORTING SERVICE DEPENDENCIES

Inference gateway (context capture). RQS / HERMES (chunk metadata).

Data produced: {exchange_id, system_prompt_tokens, retrieved_chunks[], history_tokens, user_message_tokens, total_tokens}

Data consumed: Provenance / MOIRAI chunk records. RQS / HERMES chunk metadata and match scores. ERAS / LOGOS reasoning index.

21. Disagreement Detection

Category: Session Monitoring

PURPOSE

When a query resembles one from a loaded prior session, the current response is compared against the prior response for the same query. If material divergence is detected on a specific claim or framing, an amber banner surfaces a side-by-side comparison. Dismissible but logged.

MECHANISM TARGETS

- Makes response instability visible: in standard interfaces, model response variance across sessions is completely invisible to analysts.
- Prevents tacit reliance on consistency: analysts sometimes implicitly assume that because the model said X before, X is stable.
- The side-by-side comparison raises the question of which response is correct and how that would be determined.

RUNTIME SERVICE DEPENDENCIES

- Semantic similarity service: compares current response claims against stored prior session claims for similar queries.
- Session context store: stores prior session claims when a session is loaded.

THEMIS INTEGRATION

Response divergence events are logged to TCS / MIMIR as session-level calibration signals. High divergence rate for a (interaction class, claim type) across sessions indicates calibration instability — a system-level flag distinct from analyst-level RAI. FGTS / ALETHEIA uses divergence events as correction signals when the analyst resolves the disagreement by verifying one version against a source. Provenance / MOIRAI provides the prior session claim records that make comparison possible. TVS / KAIROS is relevant when divergence results from the source evidence being superseded between sessions — a TVS validity change, not a model instability.

Active THEMIS services: Provenance/MOIRAI (C) TCS/MIMIR (P) FGTS/ALETHEIA (P) TVS/KAIROS (C)

SUPPORTING SERVICE DEPENDENCIES

Semantic similarity service. Session context store. Prior session records from Provenance / MOIRAI.

Data produced: {exchange_id, prior_session_id, similarity_score, divergence_type, dismissed:bool}

Data consumed: Provenance / MOIRAI prior session claim records. TVS / KAIROS validity change events.

22. Session Synthesis Workspace

Category: Analyst Scaffolding

PURPOSE

A dedicated right-panel workspace, separate from the Verification Queue, for the analyst own independent synthesis. Three fields: conclusions I am confident in, assumptions I am relying on, and claims I still need to verify. Structurally and visually distinct from all model output. Exportable as a session header attached to the full provenance record.

MECHANISM TARGETS

- Structural boundary between model-generated and analyst-generated content: provides a physically separate space for conclusions, preventing model framing from colonising analyst thinking.
- Peak-end rule intervention: if the analyst completes a synthesis before session close, that cognitive work becomes the memorable peak rather than the model confident final output.
- Session closure: completing the synthesis fields creates closure on open analytical questions.

RUNTIME SERVICE DEPENDENCIES

- Session context store: synthesis text fields persisted per session.

THEMIS INTEGRATION

Synthesis content is exported as the header of the Provenance / MOIRAI session record, making analyst conclusions a first-class element of the provenance chain alongside model output. TCS / MIMIR computes longitudinal correlation between synthesis conclusions and verified claims as a long-term RAI signal: do the analyst stated conclusions match the claims they actually verified? ERAS / LOGOS consumes synthesis content as a reasoning completeness signal — an analyst who synthesises

conclusions that are not supported by verified claims or ERAS-captured reasoning has a flaggable pattern.

Active THEMIS services: Provenance/MOIRAI (P) TCS/MIMIR (P) ERAS/LOGOS (C)

SUPPORTING SERVICE DEPENDENCIES

None.

Data produced: {session_id, conclusions, assumptions, to_verify, completed_at}

Data consumed: None

23. Export with Provenance Gate

Category: Output Control

PURPOSE

Export is flagged when unverified claims above a configurable confidence threshold remain in the queue. A warning count appears above the export button. Proceeding requires explicit confirmation. The exported artifact includes full provenance: model version, parameters, system prompt hash, chat mode, corpus version, session intent, all claims with verification states, retrieval trace, and a THEMIS status summary.

MECHANISM TARGETS

- Zeigarnik Effect enforced structurally: the analyst cannot close the session without explicitly acknowledging open verification items.
- Accountability at the point of consequence: export is when analysis leaves the analyst hands, making it the highest-leverage point for intervention.
- Externalising epistemic complexity: unverified claims are made visible in the export artifact rather than hidden from the downstream consumer.

RUNTIME SERVICE DEPENDENCIES

- Verification store: counts unresolved claims by confidence level.
- Provenance / MOIRAI: assembles full provenance record for export.
- Session manifest: provides parameter and configuration snapshot.

THEMIS INTEGRATION

PCES / AEGIS governs what can be exported — privilege classification determines whether any content in the session is exportable at all, and privilege log entries are included in the export artifact. PGS / NOMOS policy determines export permissions by interaction class. Provenance / MOIRAI export triggers the creation of the canonical provenance record including the full chain of custody. TCS / MIMIR contributes an RAI snapshot at export time — analyst calibration state becomes part of the analytical record. FGTS / ALETHEIA contributes the verification summary including action type distribution. ERAS / LOGOS contributes the reasoning audit record, enabling the exported work product to demonstrate that AI reasoning was complete and supported.

Active THEMIS services: PCES/AEGIS (C) PGS/NOMOS (C) Provenance/MOIRAI (P) TCS/MIMIR (C) FGTS/ALETHEIA (C) ERAS/LOGOS (C)

SUPPORTING SERVICE DEPENDENCIES

Verification store. Provenance / MOIRAI. Session manifest.

Data produced: Full session record with provenance, verification states, TCS RAI snapshot, ERAS reasoning record, PGS policy log, PCES privilege classification.

Data consumed: PCES privilege scope. PGS export policy. TCS RAI state. FGTS verification summary. ERAS reasoning index.

24. Session Manifest

Category: Transparency

PURPOSE

A modal view of the complete session configuration record: model ID, session ID, knowledge mode, RAG corpus version, temperature, top-P and top-K, system prompt hash, session intent fields, and live verification statistics. Accessible at any point during the session.

MECHANISM TARGETS

- Reproducibility: the manifest contains every parameter needed to reconstruct the conditions that produced any response in the session.
- Audit trail: a complete record of configuration, intent, and verification state at any point in time.
- Calibration awareness: showing analysts their live verification statistics mid-session provides real-time feedback on their own behavior.

RUNTIME SERVICE DEPENDENCIES

- Session context store: all manifest fields drawn from live session state.

THEMIS INTEGRATION

The Session Manifest is the Provenance / MOIRAI record for this session — it IS the provenance record rather than a summary of it. PCES / AEGIS privilege classification and matter scope are manifest fields. PGS / NOMOS policy version and active rule set are manifest fields. TCS / MIMIR contributes live RAI score and calibration flags. The manifest enables cross-session comparison by model version, parameter configuration, and policy version — making THEMIS audit queries answerable at the session level.

Active THEMIS services: PCES/AEGIS (C) PGS/NOMOS (C) Provenance/MOIRAI (B) TCS/MIMIR (C)

SUPPORTING SERVICE DEPENDENCIES

None. Manifest assembles from session context store and THEMIS service APIs.

Data produced: Session manifest record: all configuration fields plus live verification statistics plus THEMIS service status.

Data consumed: PCES privilege scope. PGS policy version. TCS live RAI score. Provenance / MOIRAI session record.

25. Compliance Gaming Detection

Category: Session Monitoring

PURPOSE

Analysts who interact with ATHENA long enough learn which behaviors satisfy the interface without requiring genuine engagement. A three-word intention gate entry clears the character minimum. Clicking dismiss on all queued items clears the export gate. Repeating the same episode commitment across sessions satisfies the checkpoint. None of these produce epistemic value, but all satisfy the current intervention logic. Without detection, behavioral telemetry flowing into TCS / MIMIR becomes corrupted with false-positive engagement signals and calibration models degrade toward meaninglessness. A gaming probability score computed by TCS / MIMIR surfaces this pattern to the analyst and to supervision before it becomes entrenched.

MECHANISM TARGETS

- Metacognitive awareness: showing analysts their own gaming pattern score activates self-regulatory behavior (Bandura). The score is framed as calibration quality, not compliance violation.
- Evaluation apprehension (Lerner and Tetlock): supervisory visibility of the gaming score is one of the strongest behavioral motivators in knowledge work — even the anticipation of review changes behavior.
- Minimum engagement floors: structural nudges that make the fast path require at least nominal engagement without blocking the action outright.

RUNTIME SERVICE DEPENDENCIES

- Session context store: holds engagement timing data and provenance panel open/close events per exchange.
- Behavioral telemetry: logs all engagement signals — gate completion times, panel open durations, queue action sequences, time between queue item display and action.
- Semantic similarity service: computes cosine similarity of intention gate entries and episode commitments against the analyst prior five sessions for the same interaction class.
- TCS / MIMIR analytics function: computes gaming probability score from four behavioral signals — intention gate semantic drift, time-on-provenance before verification action, batch dismissal pattern, and episode commitment semantic drift.

THEMIS INTEGRATION

TCS / MIMIR is the primary owner of this intervention — gaming probability score computation is a TCS analytics function extension, not a new service. TCS computes the score from four behavioral signal streams, logs it per session, incorporates it as a calibration weight modifier (high gaming score reduces the weight of verification outcomes from that session in FGTS calibration updates), and exposes it via the existing TCS calibration dashboard API. Provenance / MOIRAI logs the gaming score in the session manifest record. FGTS / ALETHEIA receives verification outcomes annotated with their gaming-adjusted confidence weight.

Active THEMIS services: Provenance/MOIRAI (P) TCS/MIMIR (B) FGTS/ALETHEIA (P)

SUPPORTING SERVICE DEPENDENCIES

Semantic similarity service (intention gate and episode commitment text comparison across sessions). Behavioral telemetry (engagement signal streams).

Data produced: {session_id, gaming_probability_score, signal_breakdown: {gate_similarity, panel_time_score, batch_dismissal_flag, commitment_drift}, computed_at}

Data consumed: Prior session intention gate text and episode commitments from Provenance / MOIRAI. Behavioral telemetry engagement signals.

26. Pressure-Aware Session Mode

Category: Session Scaffolding

PURPOSE

Filing deadlines, hearing preparation, and urgent client requests are the conditions under which AI-assisted work is most consequential and under which verification quality is most likely to degrade. The current twenty-four interventions apply identically to a low-stakes background research query and a brief due in four hours. Under time pressure, some interventions should compress to reduce friction that causes bypassing; others should harden because the stakes of an error are higher. The intervention profile should be calibrated to the declared pressure context, not uniform across all sessions.

MECHANISM TARGETS

- Stakes recalibration (Lerner and Tetlock): consequence salience must remain active throughout the session, not only at session open. Pressure mode keeps the declared deadline and deliverable visible in the persistent header bar.
- Friction calibration: compressing gates under time pressure is counterintuitive but research-supported. Forcing a time-pressured analyst through a full two-field intention gate increases the probability of circumvention. A compressed but mandatory single-field version produces lower-quality prior elicitation but higher compliance.
- Asymmetric hardening: easier in, harder out. The export gate in Deadline-Critical mode is the critical asymmetry — the entry friction reduces while the exit enforcement increases, reflecting the actual risk distribution under deadline pressure.

RUNTIME SERVICE DEPENDENCIES

- Session context store: stores declared pressure level and activates corresponding intervention profile for the duration of the session.
- PGS / NOMOS policy engine: governs deadline-critical export gate hardening rules including supervisor acknowledgment requirement for blocked exports.
- ERAS / LOGOS: provides claim density score per response to determine which response triggers the post-session reformulation requirement in Deadline-Critical mode.

THEMIS INTEGRATION

PGS / NOMOS governs the Deadline-Critical export gate hardening: any unverified HIGH-confidence claim blocks export outright and the block requires supervisor acknowledgment — this is a PGS policy expression, not an ATHENA-layer rule. Provenance / MOIRAI logs pressure mode as a session attribute, enabling TCS / MIMIR to stratify calibration separately by pressure mode. A verification action in Deadline-Critical mode carries different behavioral context than the same action in Standard mode. FGTS / ALETHEIA receives verification outcomes annotated with pressure mode for stratified ground truth analysis. ERAS / LOGOS contributes claim density scoring to determine the post-session reformulation target in Deadline-Critical mode.

Active THEMIS services: PGS/NOMOS (P) Provenance/MOIRAI (P) TCS/MIMIR (P) FGTS/ALETHEIA (P) ERAS/LOGOS (C)

SUPPORTING SERVICE DEPENDENCIES

Session context store (pressure profile activation). PGS policy engine (export gate rules). ERAS / LOGOS (claim density per response).

Data produced: {session_id, pressure_level: standard|elevated|deadline_critical, deadline, deliverable_type, profile_activated_at, post_session_reformulation_required:bool}

Data consumed: ERAS / LOGOS claim density scores. PGS deadline-critical policy rules.

27. Multi-Analyst Session Framework

Category: Session Scaffolding

PURPOSE

Legal work is collaborative. A junior associate drafts AI-assisted research and a senior partner reviews it. A matter transfers from one analyst to another mid-analysis. Two attorneys work a deposition simultaneously using ATHENA. None of these workflows exist in the current design. Each requires different interface requirements, attribution standards, and THEMIS integration — supervisory review mode for oversight workflows, matter handoff mode for continuity across analyst transitions, and collaborative session mode for simultaneous multi-analyst work.

MECHANISM TARGETS

- Attribution accuracy: per-analyst attribution for all actions (queries, verifications, annotations) is the foundational requirement. Without it, the provenance chain cannot distinguish whose judgment produced which outcome.
- Handoff acknowledgment: the receiving analyst must explicitly acknowledge each section of the handoff brief — not a single dismiss — activating prospective memory for the prior analyst's unresolved items.
- Collaborative disagreement surfacing: when two analysts in the same session take conflicting verification actions on the same claim, that conflict is explicit and visible rather than silently overwritten.

RUNTIME SERVICE DEPENDENCIES

- Session context store: manages multi-analyst session state, per-analyst query and action attribution, shared session intent, and collaborative disagreement events.
- DPS / CODEX: generates handoff briefs from prior session Provenance / MOIRAI records, stores supervisory review records, manages multi-analyst document provenance attribution.
- Provenance / MOIRAI: receives per-analyst attributed events for all three modes. The provenance graph must support multiple analyst nodes per session turn.
- PCES / AEGIS: validates analyst authorization for supervisory review access — only analysts with supervisory authorization on the matter can enter review mode for another analyst's session.

THEMIS INTEGRATION

PCES / AEGIS governs supervisory review authorization — which analysts have matter-level supervisory access is a privilege concern, not just a role concern. Provenance / MOIRAI receives significant new event types: per-analyst attributed turn records, supervisory review events (including state overrides with reasoning), and handoff brief generation records. The provenance graph schema must support multiple analyst nodes per session. TCS / MIMIR calibrates each analyst independently even within shared sessions — collaborative session mode does not merge calibration profiles. FGTS / ALETHEIA receives dual-analyst verification outcomes as high-value ground truth signals: when two analysts independently verify the same claim, that is stronger than single-analyst verification. Supervisory override of a verification state is the highest-quality ground truth signal in the system.

Active THEMIS services: PCES/AEGIS (C) Provenance/MOIRAI (P) TCS/MIMIR (C) FGTS/ALETHEIA (P)

SUPPORTING SERVICE DEPENDENCIES

DPS / CODEX (handoff brief generation, supervisory review record management). PCES / AEGIS (supervisory authorization check).

Data produced: *Supervisory review record: {session_id, reviewer_analyst_id, reviewed_analyst_id, state_overrides[], annotations_added[], review_completed_at}. Handoff brief: {matter_id, from_analyst_id, to_analyst_id, session_summary, unverified_claims[], prior_commitments[], generated_at}.*

Data consumed: *PCES supervisory authorization grants. Provenance / MOIRAI prior session records (for handoff brief generation and supervisory review). TCS / MIMIR per-analyst calibration profiles.*

28. Cross-Analyst Disagreement Detection

Category: Session Monitoring

PURPOSE

Intervention 21 covers the same analyst asking a similar question across sessions and receiving a divergent model response. The gap is different analysts on the same matter: Analyst A asks about a regulatory position and receives one characterization; Analyst B asks an overlapping question and receives a materially different one. Neither analyst sees the discrepancy. A brief assembled from both sessions may contain contradictory factual premises that no single analyst would catch because no single analyst saw both responses. The matter coordinator sees cross-analyst disagreement events in a matter-level dashboard; individual analysts receive a narrow notification without seeing the other analyst's session.

MECHANISM TARGETS

- Attribution-blind notification: neither analyst is shown the other's full session. Showing one analyst the other's response would anchor their reasoning to the prior framing, defeating the purpose of independent analysis.
- Matter-level visibility: the supervising attorney or matter coordinator sees the full cross-analyst divergence event — both queries, the diverging claims, model versions, and TVS validity state of relevant sources.
- Calibration instability signal: cross-analyst divergence on a verified claim is a qualitatively different signal from single-analyst disagreement detection. It indicates the model is producing inconsistent outputs for the same claim type on the same matter, which is a matter-level calibration flag distinct from general model drift tracked by MDS / KRONOS.

RUNTIME SERVICE DEPENDENCIES

- Semantic similarity service: compares each new query against prior queries from other analysts on the same matter, retrieved from Provenance / MOIRAI.
- Provenance / MOIRAI: provides prior analyst queries and claims for same-matter cross-session comparison via a matter-level cross-session query capability.
- Session context store: tracks matter-level analyst query fingerprints for real-time comparison on each new submission.

THEMIS INTEGRATION

Provenance / MOIRAI provides the cross-session query capability that makes this intervention possible — prior analyst queries and responses on the same matter must be retrievable at submission time. This requires a matter-level index on Provenance / MOIRAI records. TCS / MIMIR receives cross-analyst

divergence events as matter-level calibration instability signals — distinct from the general model drift that MDS / KRONOS monitors at the platform level. FGTS / ALETHEIA receives cross-analyst verification conflicts (Analyst A confirms, Analyst B disputes the same claim) as high-value ground truth signals. TVS / KAIROS provides validity context for divergence events — some apparent disagreements are explained by a source being superseded between the two sessions, which is a TVS event rather than a model instability event.

Active THEMIS services: Provenance/MOIRAI (C) TCS/MIMIR (P) FGTS/ALETHEIA (P) TVS/KAIROS (C)

SUPPORTING SERVICE DEPENDENCIES

Semantic similarity service (cross-analyst query comparison). Provenance / MOIRAI matter-level cross-session query index.

Data produced: {matter_id, analyst_a_session_id, analyst_b_session_id, query_similarity_score, diverging_claims[], divergence_type: model_instability|source_supersession|framing_difference, created_at}

Data consumed: Provenance / MOIRAI matter-level query and claim records across analysts. TVS / KAIROS source validity state at time of each session.

THEMIS Integration Matrix

C = Consumer (ATHENA intervention reads from this THEMIS service). P = Producer (ATHENA intervention writes to this THEMIS service). B = Both. Dash = No direct integration.

Intervention	PCES AEGIS	PGS NOMOS	Provenan ce MOIRAI	TCS MIMIR	FGTS ALETHEI A	TVS KAIROS	RQS HERMES	KCS ARGUS	ERAS LOGOS
1. Session Intent	P	P	-	P	-	-	-	-	-
2. Chat Mode	C	C	P	C	-	-	C	-	-
3. System Prompt	-	B	P	-	-	-	-	-	-
4. Parameters	-	-	P	C	-	-	-	-	-
5. Prompt Repo	-	C	-	P	-	-	-	-	-
6. Session History	-	-	C	C	C	-	-	-	-
7. Input Classification	-	C	P	P	-	-	-	-	C
8. Intention Gate	-	-	-	P	P	-	-	-	-
9. Claim Confidence Badges	-	-	P	C	C	-	C	-	C
10. Source Type Badges	-	-	P	P	P	C	C	-	-
11. Counter-Position	-	C	-	P	-	-	-	-	P
12. Agreement Tracker	-	-	-	P	P	-	-	-	-
13. Verification Queue	C	-	P	P	P	-	C	-	-
14. Verification Actions	-	-	P	P	P	C	-	-	-
15. Reformulation Gate	-	-	-	P	P	-	-	-	C
16. Episode Boundary	-	-	-	P	P	-	-	-	-
17. Per-Message Feedback	-	-	-	P	P	-	-	-	-
18. Annotations	-	-	P	P	P	-	-	-	C
19. Branching	-	-	P	-	P	-	-	-	-
20. Context Inspector	-	-	C	-	-	-	C	-	C
21. Disagree Detection	-	-	C	P	P	C	-	-	-
22. Synthesis Workspace	-	-	P	P	-	-	-	-	C
23. Export Gate	C	C	P	C	C	-	-	-	C
24. Session Manifest	C	C	B	C	-	-	-	-	-
25. Compliance Gaming Detection	-	-	P	B	P	-	-	-	-

Intervention	PCES AEGIS	PGS NOMOS	Provenan ce MOIRAI	TCS MIMIR	FGTS ALETHEI A	TVS KAIROS	RQS HERMES	KCS ARGUS	ERAS LOGOS
26. Pressure-Aware Session Mode	-	P	P	P	P	-	-	-	C
27. Multi-Analyst Session Framework	C	-	P	C	P	-	-	-	-
28. Cross-Analyst Disagreement Detection	-	-	C	P	P	C	-	-	-

C	Consumer: reads from THEMIS	P	Producer: writes to THEMIS	B	Both: reads and writes
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Primary Data Flows

The FGTS Calibration Loop

The most critical data flow is the calibration feedback loop. TCS / MIMIR serves calibrated confidence scores to Claim Confidence Badges. The analyst reads the response, expands the provenance panel in the Verification Queue (using RQS / HERMES source documents retrieved via Provenance / MOIRAI), and takes a typed verification action. That action flows to FGTS / ALETHEIA as a correction event. FGTS routes it to TCS to update confidence models and to RQS to flag retrieval misses. Updated TCS scores are served on the next request. This loop only closes if analysts take explicit verification actions. High Verification Queue completion rate is both a RAI input to TCS and a system health metric for FGTS.

The TVS Currency Chain

KCS / ARGUS monitors external sources and sends active invalidation events to TVS / KAIROS. TVS propagates validity score changes through the Provenance / MOIRAI DAG to all claims whose ancestry includes the invalidated source. RQS / HERMES monitors the TVS-retrieval intersection to detect cases where superseded evidence is still being returned by retrieval. Source Type Badges for Grounded claims surface the current TVS validity score so analysts can see whether a retrieved source remains current. Disagreement Detection uses TVS change events to distinguish genuine evidence supersession from model instability.

The ERAS Reasoning Audit Chain

ERAS / LOGOS captures reasoning at the turn level from every governed interaction. Claim Confidence Badges surface ERAS confidence signals for individual claims and flag unsupported claims where ERAS detects no retrieved evidence backing. Context Window Inspector displays the ERAS reasoning capture for the exchange, linking retrieved context to model claims. Analyst annotations and reformulation gate submissions are consumed by ERAS as evidence of analyst engagement with the reasoning. The Export Gate includes the ERAS reasoning record in every exported artifact, enabling professional responsibility demonstration that AI reasoning was complete, supported, and reviewed.

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